ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE IMPACTS OF FOREST AGE ON THE SPATIAL AND TEMPORAL GENERALISATION OF RANDOM FOREST EVAPOTRANSPIRATION MODELS

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ABSTRACT

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Key words: Evapotranspiration, Forest Age, Machine Learning, FLUXNET.
Word count: 4000

DEDICATION AND ACKNOWLEDGMENTS

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
August 16, 2026

CONTENTS

1 AIMS AND OBJECTIVES

2 METHODS

2.1 Data Acquisition and Setup

Flux tower data were downloaded from the FLUXNET2015 dataset [1].

2.2 Site Clustering

2.3 Random Forest modelling

In this study, Random Forest models were trained and tested using different methods outlined below.

2.3.1 Forward Feature Selection

2.3.2 Random Sampling of Training Data

$$ET_i = RF_{Mod}(T\_air_i, SW\_rad_i, W\_Speed_i, \sum_{i-14}^{i} P_i, LAI) \tag{1}$$

$$ET_i = RF_{Mod+Age}(T\_air_i, SW\_rad_i, W\_Speed_i, \sum_{i-14}^{i} P_i, LAI, Stand\_Age) \tag{2}$$

2.3.3 Time Series Decomposition

2.4 A Two Site Case Study

3 RESULTS

3.1 Cluster Analysis

Cluster analysis from Gower distances and

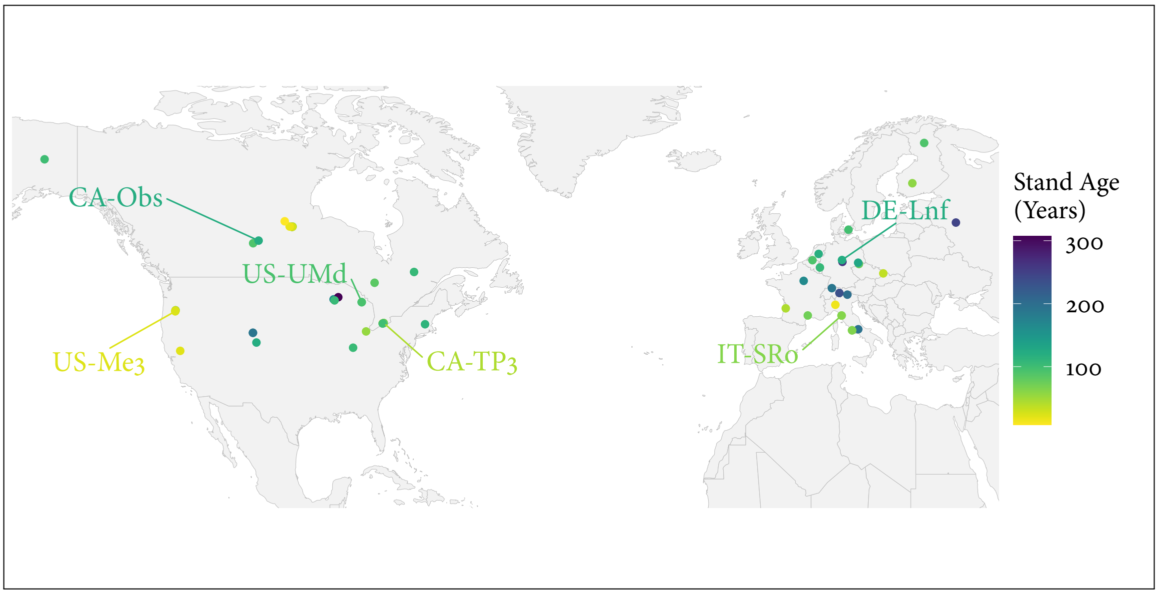

Figure 1: Map of FLUXNET sites and associated forest stand age in this study, with medoid cluster site IDs highlighted.

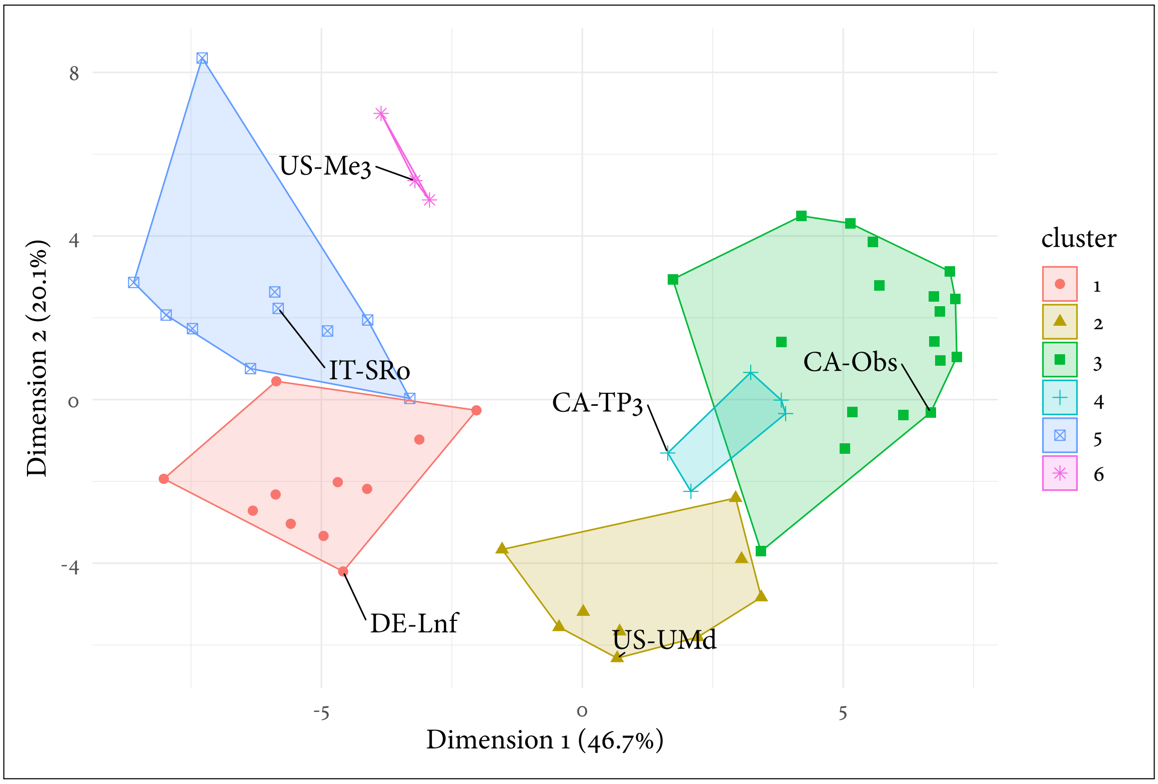

Figure 2: Cluster.

Table 1: Main characteristics of each cluster defined during cluster analysis. To see all clusters on a map see Fig. X (supplementary materials).

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

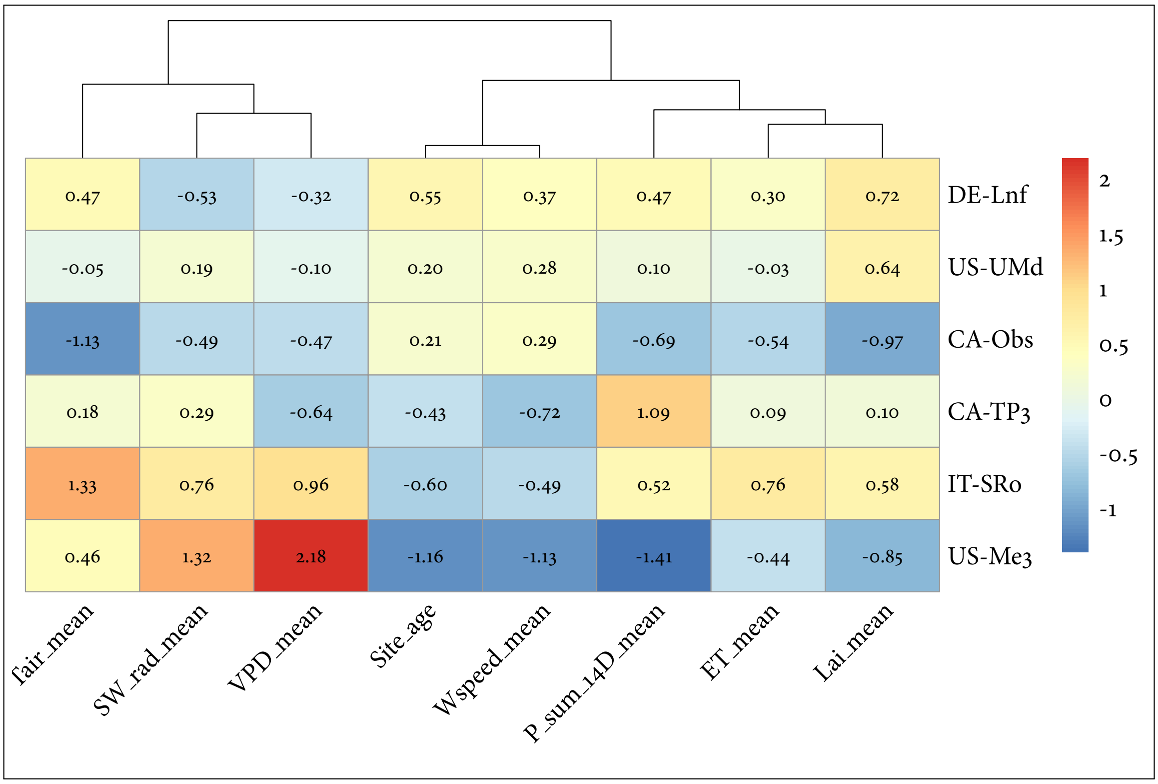

Figure 3: Heatmap of cluster medoid sites and scaled feature values. Scaled features ≥ 1.5 standard deviations from the mean were classed as 'very' (high or low, depending on direction), while features between 1 and 1.5 standard deviations from the mean were classed as high or low, depending on direction, in order to characterise clusters.

3.2 Spatial Generalisation

Suggested framing:

US-UMd (age helps) vs De-Lnf (age hurts) as case studies

For each: SHAP dependence plot for age, feature importance ranking, brief site characteristics (climate, age, cluster membership) Then the permutation test as a cross-cutting check — not per-site, but run once (or per-cluster) across the whole train/test setup to test whether age's apparent value is real signal or site-proxy leakage

Mention the other 4 clusters exist and were tested (maybe one summary table/plot of all 6 site R^2 deltas) so it's clear you didn't cherry-pick — then justify the deep dive as case studies. That's a normal, well-scoped move for a 4k-word section, not weak coverage.

Yeah, that's a sensible scope for 4k words total. Rough shape:

Spatial (bulk of results): cluster analysis overview (brief, all 6), then deep dive on DE-Tha vs CA-Obs (SHAP, feature importance, why age helps/hurts), permutation test as the credibility check Temporal: shorter section — does age still help/hurt over time at a couple of long-record sites, tying back to the leakage question (age varies within-site here, so it's a cleaner test) Discussion: synthesize — when/why age helps, leakage risk, what it means for future FLUXNET+age model building

3.2.1 Analysis 1.1

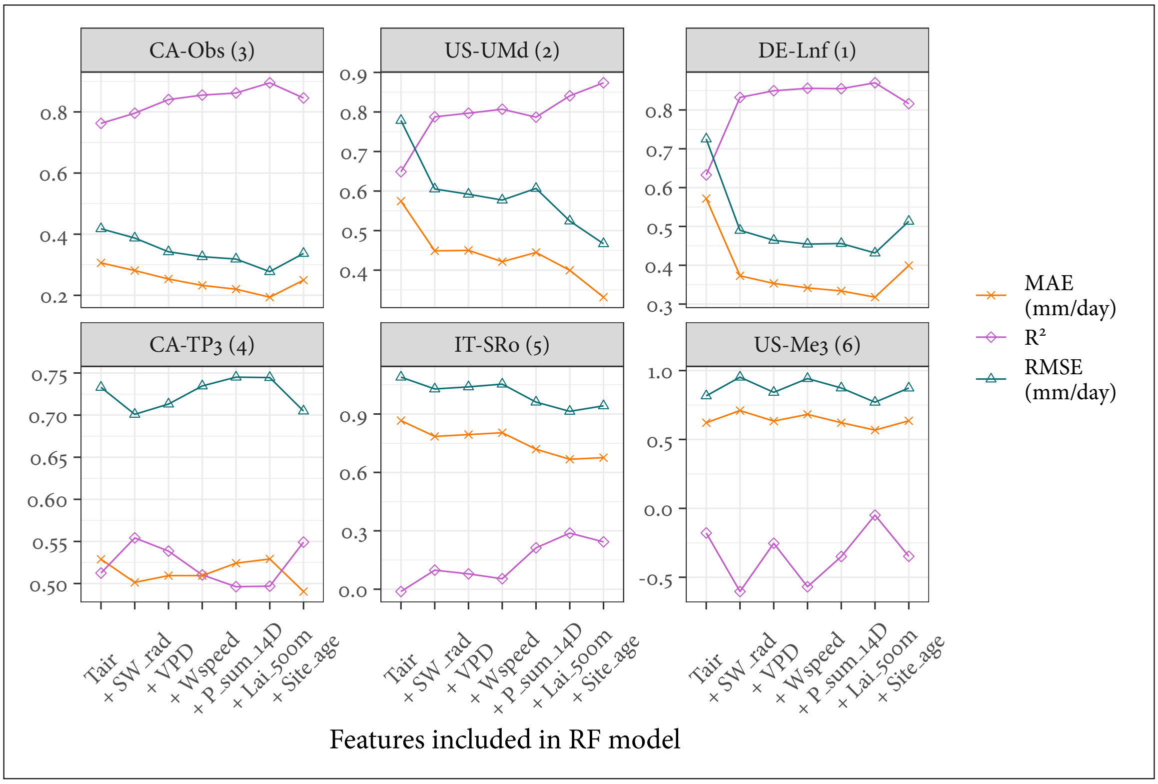

Figure 4: R^2, RMSE, and MAE for medoid cluster site predictions of ET (ordered left-right and top-bottom) by peak R^2 with clusters indicated in facet names. Models built and tested using a consecutively greater number of input features from FLUXNET and LAI data (the '+VPD model' incorporates Tair, SW_rad, and VPD as training and input features here). R^2 performance at only half of the sites was good, and also suggests that adding features up to site age improved prediction accuracy at four of the six medoid sites. Adding age as a final feature led to mixed performance across the medoid sites.

3.2.2 *Analysis 1.2*

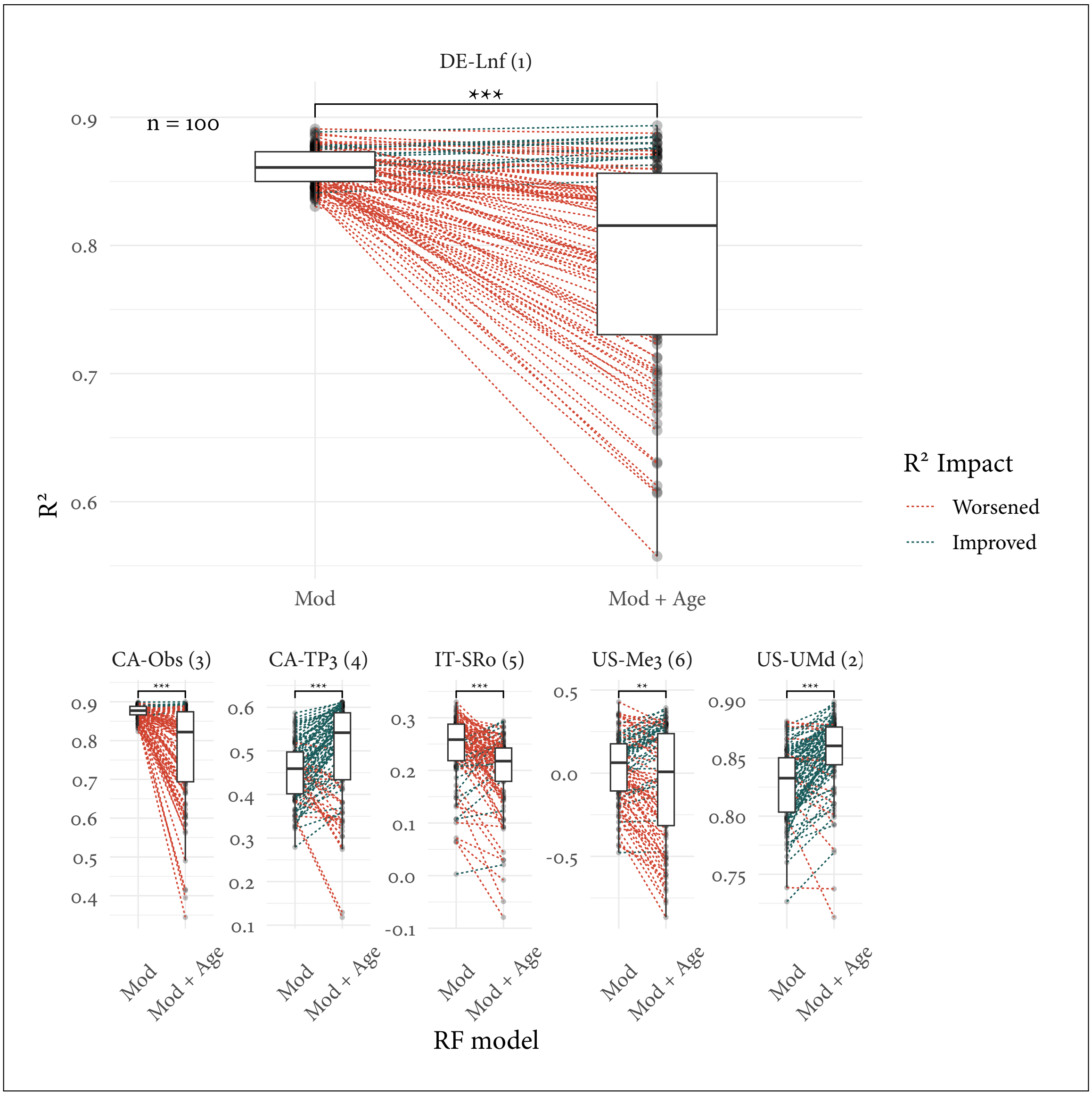

Figure 5: Paired box-plots indicate the impact on R^2 when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites. DE-Lnf is shown large for readability, with other cluster medoid sites shown at a reduced scale for comparison.

3.3 Temporal Generalisation

3.4 Case Study of US-UMd and DE-Lnf

- Look into predictions vs observed at both this R2 plot but also over timeframe
- US-UMd: Continental Site, 2715 days of data
- DE-Lnf: Temperate Site, 2670 days of data

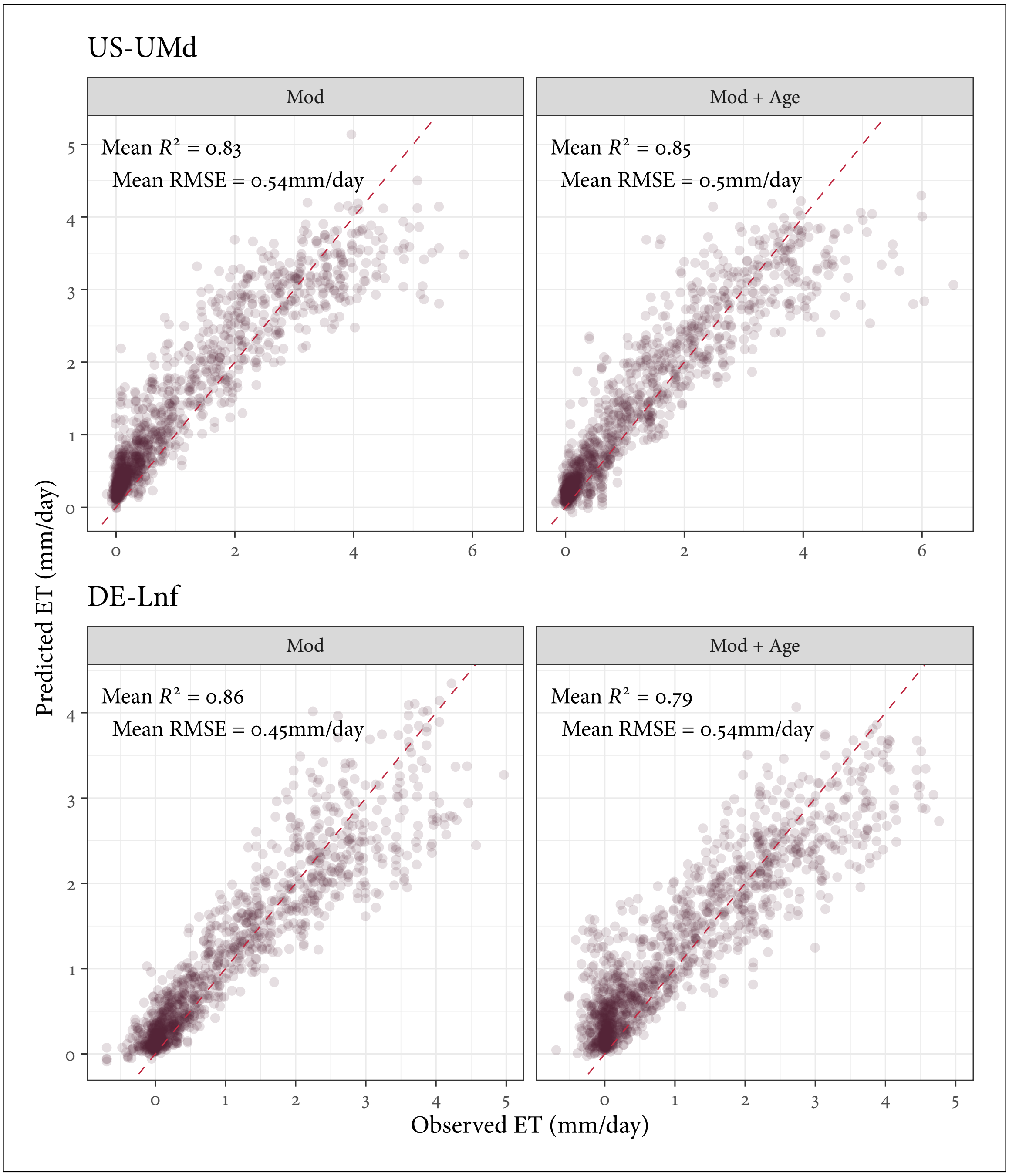
US-UMd
Mod
Mod + Age
Mean R^2 = 0.83
Mean RMSE = 0.54mm/day
Mean R^2 = 0.85
Mean RMSE = 0.5mm/day
DE-Lnf
Mod
Mod + Age
Mean R^2 = 0.86
Mean RMSE = 0.45mm/day
Mean R^2 = 0.79
Mean RMSE = 0.54mm/day
Predicted ET (mm/day)
Observed ET (mm/day)

Figure 6: P

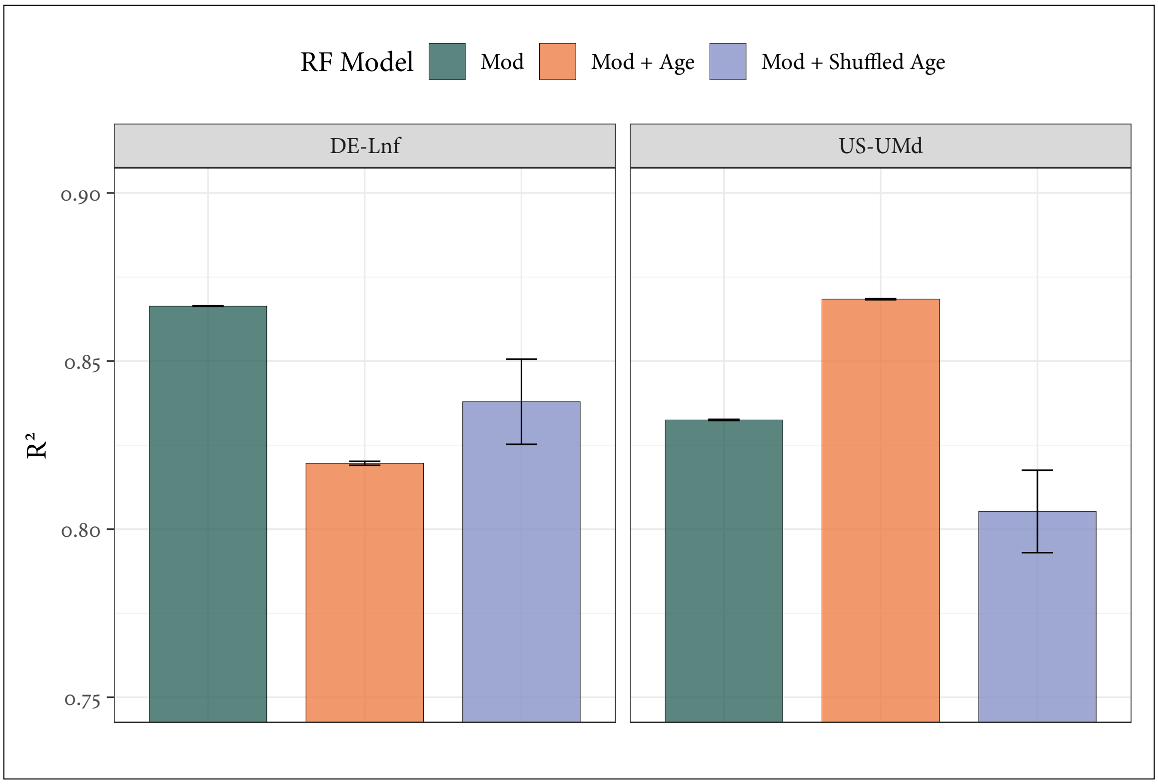

Figure 7: Bar plot showing the effect of different model setups on R^2 at DE-Lnf and US-UMd, with error bars indicating standard error. Models were trained using a leave one group out (LOGO) approach, with five repeats using different `NumPy` seeds. To assess whether site age was acting as a proxy for other site-level characteristics in these models, a model which randomly shuffled ages between sites was also formed using identical seeds and LOGO approach. This 'shuffled age' model appeared to increase R^2 at DE-Lnf compared to the Mod + Age model where an opposite effect was observed at US-UMd, suggesting that true age-ET relationships learned by the model transfer well to predicting ET at US-UMd, but not at DE-Lnf.

4 DISCUSSION

5 CONCLUSION

REFERENCES

[1] Gilberto Pastorello et al. "The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data". In: *Scientific Data* 7.1 (July 2020), p. 225. ISSN: 2052-4463. DOI: 10.1038/s41597-020-0534-3. URL: https://doi.org/10.1038/s41597-020-0534-3.

6 FIGURES AND TABLES